

Review

Regulation of the Blood–Brain Barrier by Circadian Rhythms and Sleep

Vishnu Anand Cuddapah,^{1,2} Shirley L. Zhang,² and Amita Sehgal^{2,*}

The blood–brain barrier (BBB) is an evolutionarily conserved, structural, and functional separation between circulating blood and the central nervous system (CNS). By controlling permeability into and out of the nervous system, the BBB has a critical role in the precise regulation of neural processes. Here, we review recent studies demonstrating that permeability at the BBB is dynamically controlled by circadian rhythms and sleep. An endogenous circadian rhythm in the BBB controls transporter function, regulating permeability across the BBB. In addition, sleep promotes the clearance of metabolites along the BBB, as well as endocytosis across the BBB. Finally, we highlight the implications of this regulation for diseases, including epilepsy.

The Blood–Brain Barrier Is a Dynamic Structure

A growing body of evidence demonstrates that the blood–brain barrier (BBB) is not a static structure, but instead is a site of active regulation. As a cellular structure that separates circulatory fluids (e.g., blood and hemolymph) from the brain, the BBB imposes strict regulation on molecules moving into and out of the central nervous system (CNS). Here, we focus on recent studies demonstrating that both circadian rhythms and sleep regulate the movement of molecules across and along the BBB.

Circadian rhythms are biological processes that endogenously oscillate with a period of ~24 h. This rhythm is driven by a molecular clock and can be synchronized with, but is not reliant upon, external cues (e.g., light, temperature, and feeding patterns). While the central circadian pacemaker is in the brain (e.g., the suprachiasmatic nucleus in mammals), other cells, including the cells of the BBB [1,2], have autonomous circadian rhythms driven by molecular clocks [3]. In addition, the large arteries that perfuse the brain in mammals themselves contain endogenous circadian clocks [4,5]. Critical studies originating in *Drosophila melanogaster* and then extended to mammals revealed that molecular clocks comprise transcriptional feedback loops. The core of this auto-regulatory loop in *Drosophila* involves the proteins Period (PER) and Timeless (TIM) inhibiting their own transcription at a specific time of day by acting on the transcriptional activators Clock (CLK) and Cycle (CYC) [6].

Sleep is an important behavior regulated by the circadian clock. While the timing of sleep is regulated by the circadian clock, the amount and quality of sleep depend largely on homeostatic mechanisms that ensure sufficient sleep in an organism. These two processes are separable, such that mutations in clock genes do not consistently lead to significant changes in total sleep [7], and mutations leading to decreased sleep do not disrupt the circadian clock [8,9]. As for circadian rhythms, fly models provide a robust model system to study sleep. Fly sleep recapitulates several features of sleep found in mammals, including circadian regulation, prolonged periods of quiescence with increased thresholds for arousal, electrophysiological correlates (although some of the electrophysiological signatures of sleep are specific to mammals), and changes with age [10–12].

Highlights

Recent evidence from fly models reveals an endogenous circadian rhythm at the BBB. This rhythm controls function of the permeability-glycoprotein multidrug transporter, which actively pumps both endogenous and exogenous molecules out of the CNS. Function of this transporter is decreased at night, increasing permeability to multiple substrates into the brain overnight.

Endocytosis is a newly appreciated important function for sleep. Endocytosis across the BBB is increased during sleep and, when inhibited, increases the need for sleep.

Sleep also promotes the clearance of metabolites out of the brain along paravascular spaces. During sleep, the interstitial spaces of the brain are larger, allowing for more robust movement of metabolites from interstitial spaces into the CSF. These solutes can then travel along vessels and out of the brain.

¹Division of Neurology, Children's Hospital of Philadelphia, Philadelphia, PA 19104, USA

²Center for Sleep and Circadian Neurobiology, Chronobiology Program, and Howard Hughes Medical Institute, Perelman School of Medicine at the University of Pennsylvania, Philadelphia, PA 19104, USA

*Correspondence: amita@penmedicine.upenn.edu (A. Sehgal).



In this review, we highlight recent evidence demonstrating that both circadian rhythms and sleep regulate functions of the BBB across species. Entry of hormones, cytokines, and xenobiotics from blood to brain oscillates depending on the time of day. Additionally, both the movement of metabolic products along, and endocytosis across, the BBB typically occur more robustly during sleep. Finally, we comment on how this might affect human disease and future therapeutic interventions.

***Drosophila melanogaster* Provides a Robust Model for Studies of the Blood–Brain Barrier**

The mammalian BBB consists of vascular endothelial cells connected to each other by tight junctions that form a lumen for blood flow. Endothelial cells express a set of transporters to move molecules both into and out of the brain [13], which include permeability-glycoprotein multidrug transporter [also known as Pgp, multidrug resistance protein 1, or ATP-binding cassette sub-family B member 1 (ABCB1)], an ATP-dependent efflux pump that can pump both endogenous molecules and exogenous compounds back into the lumen of blood vessels [14,15]. The surface of endothelial cells exposed to the brain is surrounded by specialized projections (endfeet) of astrocytic glial cells.

While there are differences at a gross anatomical level, the cellular structure and function of the *Drosophila* and mammalian BBB share many commonalities [16,17] (Figure 1). Instead of a closed circulatory system found in mammals, flies have an open circulatory system containing hemolymph (analogous to blood) that is separated from the CNS by two layers of glial cells: perineurial glia (PG) and subperineurial glia (SPG). The SPG layer is proximal to the CNS and forms a continuous structure with tight junctions, whereas the PG layer is proximal to the hemolymph and coated in neural lamina [18]. The SPG express Pgp-like transporters, allowing for the same active efflux that occurs in vertebrate systems [19]. Here, we discuss findings from fly and vertebrate models demonstrating how the BBB is regulated by time-of-day effects.

Circadian Regulation of Permeability through the Blood–Brain Barrier

Molecules can cross the BBB by: (i) penetration through the plasma membrane through pores or endocytosis (e.g., albumin); (ii) carrier-mediated transport systems (e.g., glucose or amino acids); (iii) direct permeation through the plasma membrane (e.g., lipophilic molecules) [20], which is

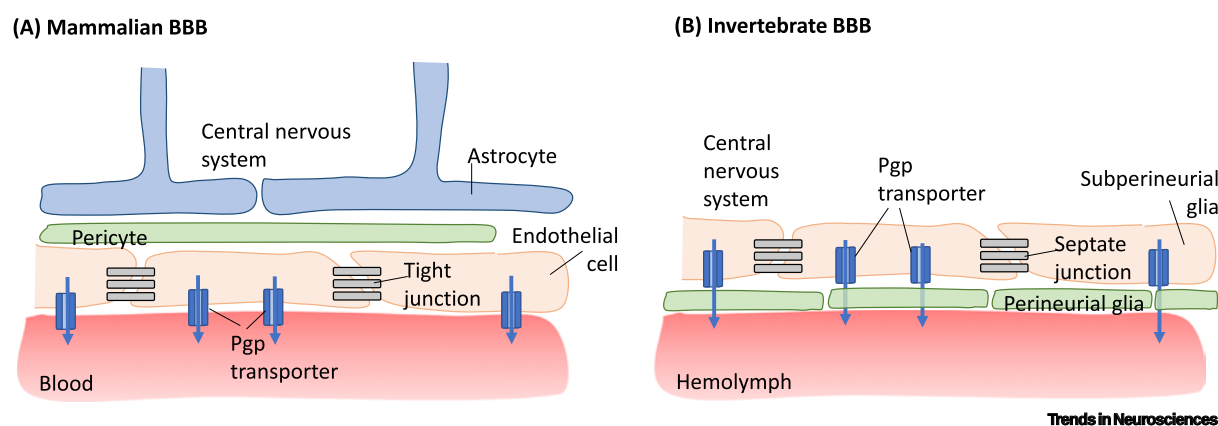


Figure 1. The Mammalian and Invertebrate Blood–Brain Barrier (BBB) Share Structural and Functional Commonalities. (A) The mammalian BBB consists of vascular endothelial cells, connected to each other by tight junctions, lining blood vessels. Endothelial cells express permeability-glycoprotein multidrug transporters (Pgp transporters) that pump both endogenous and exogenous molecules out of the central nervous system and into the blood vessel lumen. Pericytes surround endothelial cells and help regulate the barrier phenotype of the endothelial cells. Astrocytic endfeet ensheath these blood vessels. (B) The invertebrate BBB has several similar features. Hemolymph is separated from the central nervous system by two layers of glia: perineurial glia (PG) and subperineurial glia (SPG). The SPG are closer to the central nervous system and are connected by septate junctions. The SPG express Pgp transporters, analogous to mammalian vascular endothelial cells.

regulated by the activity of ATP-binding cassette (ABC) transporters; or (iv) paracellular aqueous diffusion, which is inhibited by tight junctions under normal conditions.

Rhythmic Expression of Several Secreted Molecules in the Brain

The concentrations of several molecules in the CNS undergo circadian oscillations, likely because of their rhythmic entry into the CNS (Table 1). These include tumor necrosis factor alpha (TNF α) [21], leptin [22], β -amyloid [23], delta-sleep inducing peptide (DSIP) [24], and prostaglandin D2 (PGD2) [25]. When TNF α , an inflammatory cytokine, was intravenously injected into mice with an iodine label (125 I-TNF α), uptake into the spinal cord demonstrated a circadian rhythm [21], such that greatest TNF α uptake occurred between Zeitgeber time (ZT) 20–ZT23 (ZT0, lights on; ZT12, lights off); interestingly the same rhythmic uptake did not occur in the brain [21]. Knock-down of the TNF α receptors p55 and p75 in mice eliminated entry of 125 I-TNF α into the spinal cord and brain, implicating these receptors in the uptake [26]. Disease relevance is suggested, for instance, by the finding that TNF α uptake is upregulated after spinal cord injury, even in areas of the spinal cord not directly affected by the injury [27,28].

Similarly, entry of the hunger-suppressing hormone, leptin, into the brain and spinal cord undergoes daily oscillation. Labeling of leptin with 125 I revealed greatest influx into the mouse brain and spinal cord between ZT14 and ZT18 [22]. This also occurred through a saturable transport system, because addition of unlabeled leptin inhibited the uptake of labeled leptin into the CNS [22].

β -amyloid, involved in the pathogenesis of Alzheimer's disease, also undergoes a daily oscillation in the interstitial fluid of mice, peaking between ZT18 and ZT24 [23]. Global loss of BMAL1, a component of the molecular clock, led to decreased daily oscillations of β -amyloid and increased amyloid plaque deposition [23]. This oscillation was dependent on function of the molecular clock in the suprachiasmatic nucleus (SCN) [23].

Other endogenous molecules also exhibit daily oscillations in the cerebrospinal fluid (CSF), although the mechanisms of most of these cycles remain unclear. DSIP is a peptide that promotes slow-wave sleep. While the existence of DSIP as an endogenous peptide has been called into question [29], it does appear to cross the BBB via a nonsaturable mechanism with circadian changes in brain concentration [24]. As another example, PGD2 exhibits daily oscillations in the rat CSF, peaking at 14:00 and increasing after sleep deprivation [25,30]. Injection of PGD2 into

Table 1. Timing of Peak Concentrations in the CNS of Various Endogenous and Exogenous Molecules that Undergo Circadian Oscillations

Class	Solute	Organism	Peak CNS accumulation	Animal behavior at peak	BBB mechanism	Refs
Cytokine	IL-6	Human	19:00	Active	Carrier-mediated transport	[89,90]
	TNF α	Mouse	ZT20–ZT23	Active	Carrier-mediated transport	[21,26,28]
Hormone	Leptin	Mouse	ZT14–ZT18	Active	Carrier-mediated transport	[22]
	Norepinephrine	Human	09:00–15:00	Active	Unknown	[31]
	PDG2	Rat	ZT8	Rest	Unknown	[25,30]
Peptide	β -amyloid	Mouse	ZT18–ZT24	Active	Carrier-mediated transport	[23,70]
	DSIP	Rat	ZT4–ZT8	Rest	Unknown	[24]
Xenobiotic	[18 F]MC225	Rat	ZT15	Active	Direct permeation/efflux	[35]
	Daunorubicin	Fly	ZT12	Rest	Direct permeation/efflux	[36]
	Quinidine	Rat	ZT8	Rest	Direct permeation/efflux	[34]
	Rhodamine B	Fly	ZT12–ZT16	Rest	Direct permeation/efflux	[36]

the forebrain of rats resulted in increased sleep [30]. In addition, norepinephrine [31] and interleukin 1 α (IL-1 α) [32] exhibit diurnal CNS oscillations through as yet unknown mechanisms.

Rhythmic permeability of the BBB (discussed later) is one possible mechanism to explain rhythmic expression of secreted factors in the CNS. Another possibility is circadian regulation of the interstitial space of the brain, because movement of the interstitial fluid is thought to control or modulate the passage of molecules from the brain to the vasculature. It is known that sleep increases the volume of the interstitial space [33], which conceivably leads to changes in the concentration of various molecules, although the specifics of these changes remain debated.

Daily Oscillations in Permeability of Exogenous Molecules across the BBB

Pgp regulates the permeability of many endogenous and exogenous molecules across the BBB in a circadian manner. For instance, levels of quinidine, a substrate of Pgp, were higher in the CSF and interstitial fluid of rats, as measured by microdialysis, when it was intravenously injected during their resting phase (ZT0, ZT4, and ZT8) than when they were active [34]. This suggests that Pgp is more active during periods of wakefulness.

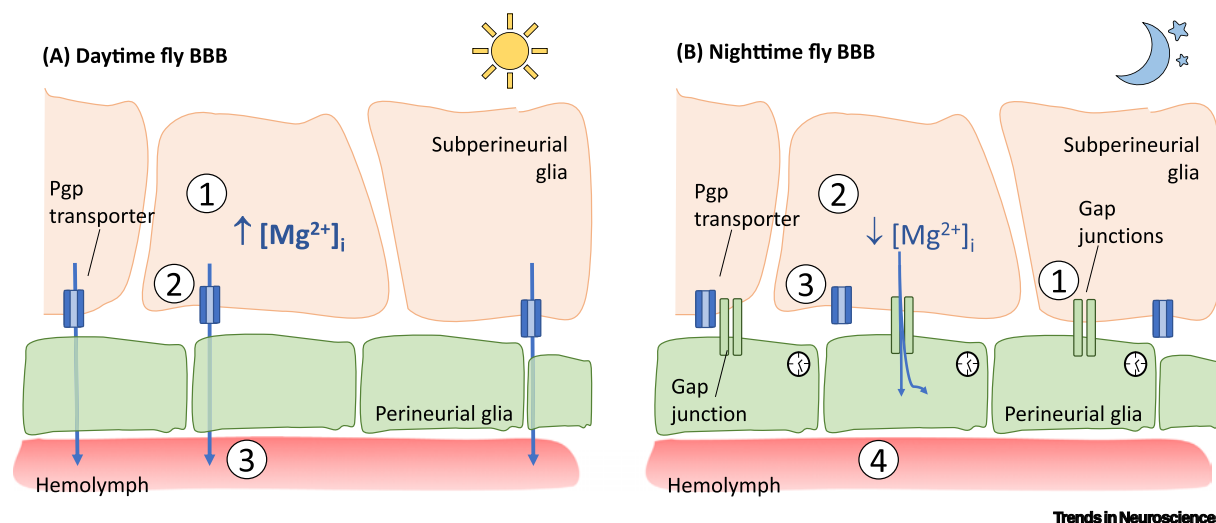
By contrast, when rats were injected with [^{18}F]MC225, a radioactive substrate of Pgp, increased accumulation in the brain occurred during the early active phase (ZT15) [35]. This suggests that the Pgp transporter was less active when rats were behaviorally active at night; however, these results are complicated by the fact that it is difficult to assess intravascular versus interstitial/CSF [^{18}F]MC225 signal when using positron emission tomography (PET) imaging. PET imaging does not allow for direct measurement of CNS concentrations and, thus, may lead to the observed discrepancy with the previous study.

As in rodents, Pgp activity is increased during times of wakefulness in flies. Injection of flies with rhodamine B, a xenobiotic, at various points throughout a day resulted in greater retention of the substrate in the brain at night (ZT12) during the rest phase [36]. This was due, in part, to active efflux through the Pgp-like transporter during the day, because genetic or pharmacological inhibition of this transporter increased accumulation of rhodamine B in the brain [36]. Loss of the circadian clock also inhibited the night-time increase in permeability, suggesting that the clock normally inhibits Pgp-like transporters at night to allow increased permeability [36]. Interestingly, genetic inhibition of Pgp transporters in flies led to increased sleep, possibly by increasing CNS permeation of peripheral sleep-promoting molecules [37].

Mechanisms Underlying Circadian Changes in BBB Permeability

In unicellular organisms and human cells, intracellular Mg^{2+} regulates the period, amplitude, and phase of circadian oscillations and is itself controlled by the molecular clock [38]. Given that Mg^{2+} influences numerous cellular processes, including protein translation and function, changes in Mg^{2+} over the course of the day can have profound effects on cell physiology.

At the fly BBB, a molecular clock in the PG cells serves to maintain oscillations of Mg^{2+} [36]. High levels of Mg^{2+} in the SPG during daytime increases the activity of Pgp transporters, thereby decreasing permeability of the BBB. Mg^{2+} increases the activity of Pgp transporters by stabilizing their binding to ATP [39]. At night (ZT14), the circadian clock in the PG upregulates expression of the gap junction proteins *Inx1* and *Inx2*, leading to increased connectivity between the SPG and PG and a decrease in Mg^{2+} in the SPG [36]. Thus, the PG function as a Mg^{2+} sink, driving intracellular Mg^{2+} down in SPG at night, which in turn decreases Pgp activity and increases permeability at the BBB [36] (Figure 2). Interestingly, glial expression of the clock protein PERIOD decreases with age in fly models, although the relevance of this finding for BBB function is not yet known [40].



Trends in Neurosciences

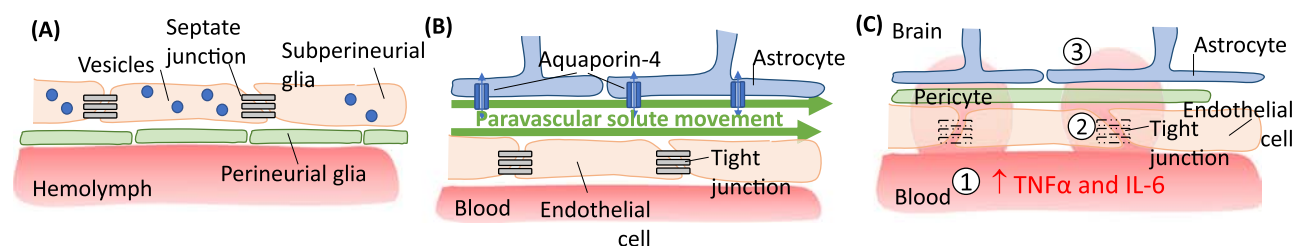
Figure 2. Circadian Regulation of Permeability-Glycoprotein (Pgp) Multidrug Transporter-Mediated Permeability across the Fly Blood-Brain Barrier (BBB). (A) During daytime, (1) the concentration of Mg^{2+} in subperineurial glia (SPG) is higher, (2) leading to increased function of Pgp transporters, thereby (3) increasing active efflux and decreasing permeability of the BBB. (B) At night-time, (1) expression of the gap junction proteins *Inx1* and *Inx2* is upregulated by the molecular clock, leading to increased connectivity between the SPG and the perineurial glia (PG). (2) The PG serve as a Mg^{2+} sink, driving intracellular Mg^{2+} down in SPG, which in turn (3) decreases Pgp activity and (4) effectively increases permeability at the BBB.

Blood-Brain Barrier Function Is Regulated by Sleep

Recent evidence demonstrates that endocytosis occurs across the BBB during sleep, and inhibition of this process increases the need for sleep [41]. Sleep regulates movement of molecules not only across the BBB, but also along it, such that clearance of waste byproducts in the perivascular space is increased during sleep [42]. Sleep loss increases permeability of the BBB through a variety of mechanisms, including increased inflammatory signaling and downregulation of tight junction proteins [43,44]. Disruption of sleep not only impairs these regulatory functions, but can also lead to breakdown of the BBB (Figure 3A–C).

Endocytic Trafficking across the BBB Is Enhanced During Sleep

Endocytosis across the glia of the BBB is a newly appreciated function for sleep. Using a fly model, endocytosis was found to be highest early during the night (ZT14), when sleep need is thought to be the highest, and lowest early during the morning (ZT2), when sleep need is low [41]. Whereas the sleep-wake state affects endocytosis, levels of endocytosis across the SPG



Trends in Neurosciences

Figure 3. Endocytosis across and Paravascular Solute Movement along the Blood-Brain Barrier (BBB) Is Increased During Sleep. (A) In fly models, endocytosis across subperineurial glia is increased in sleep. (B) In mammals, interstitial fluid enters the brain parenchyma along paravascular spaces along arteries before entering the interstitial space in the brain parenchyma, allowing for solute movement. (C) Sleep restriction leads to (1) increased activity of several proinflammatory cytokines, including tumor necrosis factor (TNF)- α and interleukin (IL)-6, which can lead to (2) downregulation of tight junction proteins and (3) increased permeability of the BBB.

of the BBB (Figure 3A) also regulate sleep. Blocking endocytosis in the SPG through the use of *Shibire*, a dominant-negative allele of dynamin, which is a GTPase involved in endocytosis and vesicle formation, increased total sleep time [41]. These findings indicate a reciprocal relationship between sleep and endocytosis, although the specific molecules trafficked in a sleep-dependent fashion are not yet known. Importantly, inhibition of endocytic trafficking did not result in overt breakdown of the BBB, because 10 kDa dextran was still restricted from the brains of these flies [41].

To identify critical components of endocytosis that drive sleep, a panel of Rab proteins was screened in flies [41]. Rab proteins are GTPases that regulate many aspects of vesicular formation, movement, and fusion. A constitutively active form of Rab11 was found to promote sleep when expressed in either the SPG or PG [41]. These data further indicate that endosomal formation and trafficking in the BBB are critical for sleep.

Mammalian models also demonstrate a correlation between sleep and vesicular trafficking across the BBB. Restriction of sleep in rats for 20 h a day over 10 consecutive days revealed a threefold increase in the number of pinocytic vesicles in CA3 hippocampal endothelial cells, as revealed by transmission electron microscopy [45]. Taken together, invertebrate and vertebrate models suggest that endocytosis and vesicular trafficking across the BBB is modulated by sleep and could represent one of the functions of sleep. Induction of prolonged wakefulness leads to increased endocytosis across the BBB at the beginning of sleep, when the need for sleep is the greatest. How this affects glial transcytosis, the movement of macromolecules from one side of the cell to the other, is unknown.

Sleep Promotes Perivascular Clearance

As mentioned earlier, in addition to regulation across the BBB, the sleep–wake state also affects the movement of molecules along the BBB. In rodent models, interstitial fluid is thought to enter the brain parenchyma along arterial perivascular spaces and exit along venous perivascular spaces [33,42] through a ‘glymphatic system’ (Figure 3B), although other studies suggest drainage of interstitial fluid along arteries and capillaries [46,47]. This system clears byproducts and waste products out of the brain and is facilitated by sleep [42]. Sleep is associated with an up to 60% increase in interstitial spaces (although controversy over the exact numbers remains), which may promote exchange between interstitial fluid and CSF due to decreased tissue resistance [42]. This CSF flux through the interstitium is facilitated by aquaporin-4 expression in astrocytes, because loss of aquaporin-4 in a mouse model decreased CSF influx and associated clearance of solutes [33]. Solute movement through perivascular and interstitial spaces is increased by the sleep state [42], but whether there is also circadian oscillation in the interstitial volume is unknown.

A well-studied toxic byproduct regularly removed from the CNS is soluble β -amyloid. As demonstrated by murine models, in healthy physiology, most β -amyloid is cleared out of the brain, while the remainder is degraded [48]. β -amyloid is cleared through multiple overlapping routes (reviewed in detail in [49]), which include substantial clearance through the BBB [50,51], as well as elimination through the interstitial fluid–CSF exchange described earlier [33]. Consistent with the role of glial aquaporin-4 in solute movement, loss of aquaporin-4 in murine models leads to impaired β -amyloid clearance from the brain [33]. β -amyloid in the CSF likely exits through cervical lymphatic vessels into cervical lymph nodes [52].

These findings are important from a disease perspective, because perivascular clearance is reduced in the aging brain, leading to accumulation of β -amyloid associated with Alzheimer’s disease. Comparison of 2–3-month-old wild-type mice with 18–20-month-old mice showed a 40% decrease in β -amyloid clearance [53]. The age-related decline was attributed to decreased arterial pulsatility and

decreased localization of aquaporin-4 to perivascular endfeet [53]. This effect may be further exacerbated by loss of sleep, given that older individuals sleep less and have decreased quality of sleep [54]. β -amyloid was found to accumulate in interstitial fluid during wakefulness and after sleep deprivation, even after just 1 night of sleep deprivation [55,56]. Similarly, *tau*, a cytoplasmic protein also associated with neurodegeneration, increases in interstitial fluid and CSF during wakefulness and after sleep deprivation in both mice and humans [57]. Thus, sleep may be restorative in neurodegenerative disease by allowing for perivascular clearance.

Sleep Loss Leads to Increased BBB Permeability

Many models demonstrate that sleep loss is associated with increased paracellular aqueous permeability, which is typically limited by tight junctions, at the BBB. When sleep was restricted in rats for 10 days, BBB permeability increased to allow the influx of markers such as fluorescein-sodium and Evans blue [58,59]. Even short periods of recovery sleep, ranging from 40 min to 120 min led to normalization of permeability, except in the hippocampus and cerebellum [45]. Similarly, sleep restriction for 6 days in mice led to increased permeability across the BBB to fluorescein and biotin, which normalized after 24 h of recovery sleep [43].

The increased BBB permeability associated with sleep deprivation has been attributed to inflammatory signaling leading to downregulation of endothelial tight junction proteins (Figure 3C). Sleep restriction leads to increased activity of several proinflammatory mediators, including C-reactive protein [60], IL-1 β [61], IL-6 [62], IL-17 [63], interferon- γ (IFN γ) [64], and TNF α [62,64–67]. IL-1 β suppresses astrocytic maintenance of the BBB, resulting in increased permeability [61]. TNF α and IL-6 lead to decreased expression and mislocalization of zonula occludens-1 (ZO-1), a protein found in tight junctions, in human brain microvascular endothelial cells [44]. Sleep restriction in rats resulted in the formation of clefts between endothelial cells in the hippocampus, as well as decreased expression of tight junction components claudin-5 and occludin [58,59]. Sleep restriction in mice also decreased the expression of tight junction proteins, including occludin, claudin-1, claudin-5, and ZO-2 [43]. These effects of cytokine signaling on BBB permeability are not specific to sleep loss, because other models that enhance TNF α and IL-6 signaling can produce a similar breakdown [68,69]. The effects of sleep loss on disruption of the BBB may be more apparent in aged mice compared with young mice [65].

Possible Roles for Circadian- and Sleep-Mediated Regulation of the Blood–Brain Barrier

The recent studies discussed earlier suggest a model in which the energy-intensive process of moving molecules into and out of the brain is compartmentalized to certain times of the day. One could speculate that, in part, the selectivity serves to distribute energy expenditure throughout the day. While we have focused primarily on recent discoveries involving the Pgp transporter and endocytosis across the BBB, there are other mechanisms of energy-dependent transport across the BBB, which involve other members of the ABC family of transporters as well the solute carrier (SLC) family of transporters. However, whether these larger superfamilies of transporters are regulated by circadian rhythms and sleep remains to be determined.

Possible Compartmentalization of Energy-Intensive Processes

Both fly and rodent models suggest that the ATP-dependent Pgp transporter is more active during periods of activity [34,36]. The transporter has an important role in limiting exposure to exogenous molecules by actively pumping these molecules from the brain parenchyma into the lumen of blood vessels. This could be advantageous from an evolutionary perspective. As animals explore their environments, they encounter possibly neurotoxic molecules and rely on active transport of these toxins out of the brain. These types of exposure would be fewer during periods of rest and inactivity, thereby decreasing the need for energy-intensive transport.

Conversely, during times of sleep, mechanisms that remove metabolic waste products are upregulated. Endocytosis across the fly BBB is upregulated during sleep and, when inhibited, increases the need for sleep [41]. Movement of solutes along perivascular spaces also has an important role in clearing potentially toxic waste products out of the brain [42]. It appears plausible that at least some of these still-unidentified waste products may also be somnogenic (sleep-inducing) compounds, so that they would increase sleep to promote clearance.

This temporal compartmentalization of the activities of the BBB, with active efflux primarily occurring during the day and endocytosis with perivascular clearance occurring mostly during the night, may conserve energy over a day–night cycle. However, these processes are not mutually exclusive. For example, Pgp transporters, which are under circadian control, also have a role in the clearance of β -amyloid from the brain [70]. Thus, while the absolute levels of Pgp-related transport or endocytosis across the BBB change, these processes would not be expected to be turned completely ‘on’ or ‘off’ at any time of day. In addition, sleep deprivation has multiple effects that likely affect the balance of active efflux during wake and endocytosis during sleep. For example, while the concentration of cytokines in the CSF may be regulated by circadian changes in permeability across the BBB [21,22], these same cytokines may accumulate during sleep deprivation to significantly increase BBB permeability.

The Impact on Disease: Do Changes in BBB Permeability Affect Seizures?

Disruption of circadian- and sleep-mediated regulation of the BBB could contribute to disease. A comprehensive discussion of possible disease relevance is beyond the scope of the current article, and we focus on epilepsy, where converging data appear to point at a prominent sleep and/or circadian disease component, possibly acting via regulation of the BBB. Postmortem hippocampi from patients with epilepsy demonstrate breakdown of the BBB with increased extravasation of serum components into brain parenchyma [71,72]. Preliminary data also indicate that acute breakdown of the BBB can cause seizures in a porcine model and in patients undergoing BBB disruption for improved delivery of chemotherapy to the brain [73]. These studies indicate that loss of BBB integrity could worsen seizure burden.

Epilepsy is linked to circadian-dependent processes. For instance, seizure occurrence follows a cyclic pattern [74,75]. A study of 131 adult patients with epilepsy using intracranial electroencephalography (EEG) recording demonstrated a peak in seizures between 16:00 and 19:00 for seizures originating from occipital or mesial temporal cortex, and between 04:00 and 07:00 for seizures from the frontal or parietal lobes [76]. Similarly, a study of 225 children with epilepsy demonstrated a temporal distribution to seizures, with temporal lobe seizures occurring between 21:00 and 09:00, frontal lobe seizures between 00:00 and 06:00, parietal lobe seizures between 06:00 and 09:00, and occipital lobe seizures between 09:00 and 12:00 [77]. Why there are differences in the temporal distribution of seizures between adults and children is not known, but could reflect the diverse nature of epilepsy syndromes that occur at different ages, (e.g., Panayiotopoulos syndrome typically occurs in childhood and juvenile myoclonic epilepsy typically occurs in adolescence). Whether this periodicity in seizure frequency is related to BBB dysfunction is also not well understood. Nevertheless, circadian disruption of the BBB is likely not the only way circadian dysregulation increases susceptibility to seizures. For example, global disruption of BMAL1 or specific disruption of CLOCK, both components of the molecular clock, in neurons led to a lower seizure threshold in mouse models [78,79].

Additionally, sleep loss has been demonstrated to increase seizure frequency in both animal [80,81] and human [82,83] studies and to increase the concentration of excitatory neurotransmitters [84]. As detailed earlier, sleep deprivation is associated with a breakdown of the BBB, possibly through an inflammatory mechanism. Whether sleep-dependent regulation at the BBB, including sleep-

related perivascular solute clearance and endocytosis across the BBB, contributes to seizure burden is unknown.

A Role for Chronotherapy Based on Circadian Changes in the BBB

Circadian changes in BBB permeability could be leveraged to improve disease therapy. In a fly model of epilepsy, administration of the antiseizure medication phenytoin at night, when Pgp-mediated efflux is low, was more effective in decreasing seizure burden compared with daytime administration [36]. This effect was dependent on the molecular clock, because inhibition of CYCLE, a *per* transcriptional activator, in the BBB blocked the circadian responsiveness to phenytoin [36]. These results likely extend to patients with epilepsy, as demonstrated in a cohort of 103 patients who were being treated with phenytoin and carbamazepine [85]. When all or most of these medications were given at 20:00, as opposed to a more typical dosing schedule of equal doses twice daily, there was significant improvement in seizure control with decreased adverse effects and maintenance of appropriate drug levels in blood [85]. Thus, a 'chronotherapeutic' dosing schedule could allow for improved efficacy at lower dosages, thereby decreasing the likelihood of toxic adverse effects.

Other chronotherapeutic considerations may include giving a medication when it is optimally metabolized or when the symptoms are typically high. In a patient with epilepsy, this would involve dosing a medication when seizure frequency is the greatest [86]. For example, in a study of 17 children with nocturnal and early-morning seizures, 15 children had >50% reduction in seizure frequency when their night-time dose of antiseizure medications was made proportionally larger [87].

Concluding Remarks and Future Perspectives

The BBB is dynamically regulated by both circadian rhythms and sleep. In the fly, circadian changes in Pgp transporter activity controls permeability of serum components into the brain. Additionally, in mammals, sleep is associated with increased perivascular clearance of waste metabolites out of the brain, and in flies with increased endocytosis across the BBB. Despite progress in understanding the regulation of the BBB, many questions remain (see Outstanding Questions). Identification of the critical molecules that traffic into and out of the brain continues to be important. Identification of somnogenic molecules that arise in the periphery and cross the BBB to accumulate in the awake brain will further elucidate the physiological control of sleep. Understanding how sleep deprivation might affect circadian regulation will also likely yield important insights. Sleep restriction has wide-ranging effects, including fundamental changes, such as decreased glucose uptake into the brain [43].

Sleep and circadian clocks are closely linked to human neurological disease. Among the clearest examples is epilepsy, which demonstrates a circadian rhythm of seizure occurrence. Neurodegenerative diseases, including Alzheimer's disease and Parkinson's disease, have also been associated with sleep and circadian disturbances, as reviewed elsewhere [88]. In addition to understanding disease pathology, circadian regulation is relevant for the treatment of disease. Chronotherapeutic approaches that capitalize on times and states of increased BBB permeability have the potential to improve patient outcomes while minimizing adverse effects. Thus, understanding the role of sleep- and circadian-related changes in BBB has the potential to improve the management of human disease.

Acknowledgments

This work was supported by NIH grant R37NS048471 to A.S. and by the Howard Hughes Medical Institute (A.S.).

References

1. Carver, K.A. *et al.* (2014) Rhythmic expression of cytochrome P450 epoxigenases CYP4x1 and CYP2c11 in the rat brain and vasculature. *Am. J. Physiol. Cell Physiol.* 307, 989
2. Gao, Y. *et al.* (2014) Clock upregulates intercellular adhesion molecule-1 expression and promotes mononuclear cells adhesion to endothelial cells. *Biochem. Biophys. Res. Commun.* 443, 586–591

Outstanding Questions

Does sleep deprivation affect the circadian rhythm of BBB permeability? As recently demonstrated in the fly BBB, permeability at the BBB increases because of decreased Pgp activity at night. Understanding how sleep loss, common in modern society, might modulate Pgp transporter activity may shed light on essential functions of sleep.

What are the molecules that accumulate in the awake brain to promote sleep?

What molecules are trafficked across the BBB by endocytosis during sleep? While recent data from the fly BBB highlight the importance of endocytosis during sleep, further studies are necessary to identify the critical molecules being trafficked.

What is the physiological relevance of changing molecular transport across the BBB depending on the time of day or state of an animal? In this article, we hypothesize that time of day or behavioral state temporally compartmentalizes different types of energy-intensive molecular trafficking.

Is sleep- and circadian-regulated permeability of the BBB relevant for human disease? Disrupted sleep and circadian rhythms may contribute to a variety of neurological diseases, including epilepsy. Further studies investigating whether sleep- and circadian-regulated permeability of the BBB is disrupted in disease states could open new avenues of treatment.

How can medication dosing schedules be optimized to account for circadian changes in BBB permeability, for instance in epilepsy? Given that numerous clinical studies demonstrate a clear circadian rhythm in seizure frequency in patients with epilepsy, it is possible that leveraging time-related changes in BBB permeability could allow for more effective seizure treatment with fewer off-target adverse effects.

3. Mohawk, J.A. *et al.* (2012) Central and peripheral circadian clocks in mammals. *Annu. Rev. Neurosci.* 35, 445–462
4. Davidson, A.J. *et al.* (2005) Cardiovascular tissues contain independent circadian clocks. *Clin. Exp. Hypertens.* 27, 307–311
5. Durgan, D.J. *et al.* (2017) The rat cerebral vasculature exhibits time-of-day-dependent oscillations in circadian clock genes and vascular function that are attenuated following obstructive sleep apnea. *J. Cereb. Blood Flow Metab.* 37, 2806–2819
6. Dubowy, C. and Sehgal, A. (2017) Circadian rhythms and sleep in *Drosophila melanogaster*. *Genetics* 205, 1373–1397
7. Hendricks, J.C. *et al.* (2003) Gender dimorphism in the role of cycle (BMAL1) in rest, rest regulation, and longevity in *Drosophila melanogaster*. *J. Biol. Rhythm.* 18, 12–25
8. Koh, K. *et al.* (2008) Identification of SLEEPLESS, a sleep-promoting factor. *Science* 321, 372–376
9. Shi, M. *et al.* (2014) Identification of Redeye, a new sleep-regulating protein whose expression is modulated by sleep amount. *eLife* 3, e01473
10. Shaw, P.J. *et al.* (2000) Correlates of sleep and waking in *Drosophila melanogaster*. *Science* 287, 1834–1837
11. Nitz, D.A. *et al.* (2002) Electrophysiological correlates of rest and activity in *Drosophila melanogaster*. *Curr. Biol.* 12, 1934–1940
12. Cirelli, C. and Bushey, D. (2008) Sleep and wakefulness in *Drosophila melanogaster*. *Ann. N. Y. Acad. Sci.* 1129, 323–329
13. Daneman, R. and Prat, A. (2015) The blood–brain barrier. *Cold Spring Harb. Perspect. Biol.* 7, a020412
14. King, M. *et al.* (2001) Transport of opioids from the brain to the periphery by P-glycoprotein: peripheral actions of central drugs. *Nat. Neurosci.* 4, 268–274
15. Loscher, W. and Potschka, H. (2005) Blood–brain barrier active efflux transporters: ATP-binding cassette gene family. *NeuroRx* 2, 86–98
16. Hindle, S.J. and Bainton, R.J. (2014) Barrier mechanisms in the *Drosophila* blood–brain barrier. *Front. Neurosci.* 8, 414
17. DeSalvo, M.K. *et al.* (2011) Physiologic and anatomic characterization of the brain surface glia barrier of *Drosophila*. *Glia* 59, 1322–1340
18. Stork, T. *et al.* (2008) Organization and function of the blood–brain barrier in *Drosophila*. *J. Neurosci.* 28, 587–597
19. Mayer, F. *et al.* (2009) Evolutionary conservation of vertebrate blood–brain barrier chemoprotective mechanisms in *Drosophila*. *J. Neurosci.* 29, 3538–3550
20. Banks, W.A. and Kastin, A.J. (1985) Permeability of the blood–brain barrier to neuropeptides: the case for penetration. *Psychoneuroendocrinology* 10, 385–399
21. Pan, W. *et al.* (2002) Selected contribution: circadian rhythm of tumor necrosis factor- α uptake into mouse spinal cord. *J. Appl. Physiol.* 92, 62 discussion 1356
22. Pan, W. and Kastin, A.J. (2001) Diurnal variation of leptin entry from blood to brain involving partial saturation of the transport system. *Life Sci.* 68, 2705–2714
23. Kress, G.J. *et al.* (2018) Regulation of amyloid- β dynamics and pathology by the circadian clock. *J. Exp. Med.* 215, 1059–1068
24. Banks, W.A. *et al.* (1985) Modulation of immunoactive levels of DSIP and blood–brain permeability by lighting and diurnal rhythm. *J. Neurosci. Res.* 14, 347–355
25. Pandey, H.P. *et al.* (1995) Concentration of prostaglandin D2 in cerebrospinal fluid exhibits a circadian alteration in conscious rats. *Biochem. Mol. Biol. Int.* 37, 431–437
26. Pan, W. and Kastin, A.J. (2002) TNF α transport across the blood–brain barrier is abolished in receptor knockout mice. *Exp. Neurol.* 174, 193–200
27. Pan, W. *et al.* (1997) Blood–brain barrier permeability to ebratide and TNF in acute spinal cord injury. *Exp. Neurol.* 146, 367–373
28. Pan, W. *et al.* (1999) Upregulation of tumor necrosis factor α transport across the blood–brain barrier after acute compressive spinal cord injury. *J. Neurosci.* 19, 3649–3655
29. Kovalzon, V.M. and Strekalova, T.V. (2006) Delta sleep-inducing peptide (DSIP): a still unresolved riddle. *J. Neurochem.* 97, 303–309
30. Ram, A. *et al.* (1997) CSF levels of prostaglandins, especially the level of prostaglandin D2, are correlated with increasing propensity towards sleep in rats. *Brain Res.* 751, 81–89
31. Ziegler, M.C. *et al.* (1976) Circadian rhythm in cerebrospinal fluid noradrenaline of man and monkey. *Nature* 264, 656–658
32. Banks, W.A. *et al.* (1998) Diurnal uptake of circulating interleukin-1 α by brain, spinal cord, testis and muscle. *Neuroimmunomodulation* 5, 36–41
33. Iliff, J.J. *et al.* (2012) A paravascular pathway facilitates CSF flow through the brain parenchyma and the clearance of interstitial solutes, including amyloid β . *Sci. Transl. Med.* 4, 147ra111
34. Kervezee, L. *et al.* (2014) Diurnal variation in P-glycoprotein-mediated transport and cerebrospinal fluid turnover in the brain. *AAPS J.* 16, 1029–1037
35. Savolainen, H. *et al.* (2016) P-glycoprotein function in the rodent brain displays a daily rhythm, a quantitative in vivo PET study. *AAPS J.* 18, 1524–1531
36. Zhang, S.L. *et al.* (2018) A circadian clock in the blood–brain barrier regulates xenobiotic efflux. *Cell* 173, 139
37. Hindle, S.J. *et al.* (2017) Evolutionarily conserved roles for blood–brain barrier xenobiotic transporters in endogenous steroid partitioning and behavior. *Cell Rep.* 21, 1304–1316
38. Feeney, K.A. *et al.* (2016) Daily magnesium fluxes regulate cellular timekeeping and energy balance. *Nature* 532, 375–379
39. Booth, C.L. *et al.* (2000) Analysis of the properties of the N-terminal nucleotide-binding domain of human P-glycoprotein. *Biochemistry* 39, 5518–5526
40. Long, D.M. and Giebultowicz, J.M. (2018) Age-related changes in the expression of the circadian clock protein PERIOD in *Drosophila* glial cells. *Front. Physiol.* 8, 1131
41. Artiushin, G. *et al.* (2018) Endocytosis at the *Drosophila* blood–brain barrier as a function for sleep. *eLife* 7, e43326
42. Xie, L. *et al.* (2013) Sleep drives metabolite clearance from the adult brain. *Science* 342, 373–377
43. He, J. *et al.* (2014) Sleep restriction impairs blood–brain barrier function. *J. Neurosci.* 34, 14697–14706
44. Rochford, K.D. and Cummins, P.M. (2015) Cytokine-mediated dysregulation of zonula occludens-1 properties in human brain microvascular endothelium. *Microvasc. Res.* 100, 48–53
45. Gomez-Gonzalez, B. *et al.* (2013) REM sleep loss and recovery regulates blood–brain barrier function. *Curr. Neurovasc. Res.* 10, 197–207
46. Arbel-Ornath, M. *et al.* (2013) Interstitial fluid drainage is impaired in ischemic stroke and Alzheimer's disease mouse models. *Acta Neuropathol.* 126, 353–364
47. Szentistvanyi, I. *et al.* (1984) Drainage of interstitial fluid from different regions of rat brain. *Am. J. Phys.* 246, 835
48. Qosa, H. *et al.* (2014) Differences in amyloid- β clearance across mouse and human blood–brain barrier models: kinetic analysis and mechanistic modeling. *Neuropharmacology* 79, 668–678
49. Tarasoff-Conway, J.M. *et al.* (2015) Clearance systems in the brain—implications for Alzheimer disease. *Nat. Rev. Neurol.* 11, 457–470
50. Shibata, M. *et al.* (2000) Clearance of Alzheimer's amyloid-ss(1–40) peptide from brain by LDL receptor-related protein-1 at the blood–brain barrier. *J. Clin. Invest.* 106, 1489–1499
51. Storck, S.E. *et al.* (2016) Endothelial LRP1 transports amyloid- β (1–42) across the blood–brain barrier. *J. Clin. Invest.* 126, 123–136
52. Pappolla, M. *et al.* (2014) Evidence for lymphatic A β clearance in Alzheimer's transgenic mice. *Neurobiol. Dis.* 71, 215–219
53. Kress, B.T. *et al.* (2014) Impairment of paravascular clearance pathways in the aging brain. *Ann. Neurol.* 76, 845–861
54. Ohayon, M.M. *et al.* (2004) Meta-analysis of quantitative sleep parameters from childhood to old age in healthy individuals: developing normative sleep values across the human lifespan. *Sleep* 27, 1255–1273
55. Shokri-Kojori, E. *et al.* (2018) β -Amyloid accumulation in the human brain after one night of sleep deprivation. *Proc. Natl. Acad. Sci. U. S. A.* 115, 4483–4488
56. Kang, J.E. *et al.* (2009) Amyloid- β dynamics are regulated by orexin and the sleep–wake cycle. *Science* 326, 1005–1007
57. Holth, J.K. *et al.* (2019) The sleep–wake cycle regulates brain interstitial fluid tau in mice and CSF tau in humans. *Science* 363, 880–884

58. Hurtado-Alvarado, G. *et al.* (2017) Chronic sleep restriction disrupts interendothelial junctions in the hippocampus and increases blood-brain barrier permeability. *J. Microsc.* 268, 28–38
59. Hurtado-Alvarado, G. *et al.* (2016) A2A adenosine receptor antagonism reverts the blood-brain barrier dysfunction induced by sleep restriction. *PLoS One* 11, e0167236
60. Meier-Ewert, H.K. *et al.* (2004) Effect of sleep loss on C-reactive protein, an inflammatory marker of cardiovascular risk. *J. Am. Coll. Cardiol.* 43, 678–683
61. Wang, Y. *et al.* (2014) Interleukin-1 β induces blood-brain barrier disruption by downregulating Sonic hedgehog in astrocytes. *PLoS One* 9, e110024
62. Venancio, D.P. and Suchecki, D. (2015) Prolonged REM sleep restriction induces metabolic syndrome-related changes: mediation by pro-inflammatory cytokines. *Brain Behav. Immun.* 47, 109–117
63. Yehuda, S. *et al.* (2009) REM sleep deprivation in rats results in inflammation and interleukin-17 elevation. *J. Interf. Cytokine Res.* 29, 393–398
64. Hurtado-Alvarado, G. *et al.* (2018) The yin/yang of inflammatory status: blood-brain barrier regulation during sleep. *Brain Behav. Immun.* 69, 154–166
65. Opp, M.R. *et al.* (2015) Sleep fragmentation and sepsis differentially impact blood-brain barrier integrity and transport of tumor necrosis factor- α in aging. *Brain Behav. Immun.* 50, 259–265
66. Ramesh, V. *et al.* (2012) Disrupted sleep without sleep curtailment induces sleepiness and cognitive dysfunction via the tumor necrosis factor- α pathway. *J. Neuroinflammation* 9, 91
67. Chennaoui, M. *et al.* (2011) Effect of one night of sleep loss on changes in tumor necrosis factor α (TNF- α) levels in healthy men. *Cytokine* 56, 318–324
68. Silwedel, C. and Forster, C. (2006) Differential susceptibility of cerebral and cerebellar murine brain microvascular endothelial cells to loss of barrier properties in response to inflammatory stimuli. *J. Neuroimmunol.* 179, 37–45
69. Farkas, G. *et al.* (1998) Experimental acute pancreatitis results in increased blood-brain barrier permeability in the rat: a potential role for tumor necrosis factor and interleukin 6. *Neurosci. Lett.* 242, 147–150
70. Cirrito, J.R. *et al.* (2005) P-glycoprotein deficiency at the blood-brain barrier increases amyloid- β deposition in an Alzheimer disease mouse model. *J. Clin. Invest.* 115, 3285–3290
71. Rigau, V. *et al.* (2007) Angiogenesis is associated with blood-brain barrier permeability in temporal lobe epilepsy. *Brain* 130, 1942–1956
72. van Vliet, E.A. *et al.* (2007) Blood-brain barrier leakage may lead to progression of temporal lobe epilepsy. *Brain* 130, 521–534
73. Marchi, N. *et al.* (2007) Seizure-promoting effect of blood-brain barrier disruption. *Epilepsia* 48, 732–742
74. Sanchez Fernandez, I. and Loddenkemper, T. (2018) Chronotherapeutic implications of cyclic seizure patterns. *Nat. Rev. Neurol.* 14, 696–697
75. Karoly, P.J. *et al.* (2018) Circadian and circaseptan rhythms in human epilepsy: a retrospective cohort study. *Lancet Neurol.* 17, 977–985
76. Durazzo, T.S. *et al.* (2008) Temporal distributions of seizure occurrence from various epileptogenic regions. *Neurology* 70, 1265–1271
77. Loddenkemper, T. *et al.* (2011) Circadian patterns of pediatric seizures. *Neurology* 76, 145–153
78. Li, P. *et al.* (2017) Loss of CLOCK results in dysfunction of brain circuits underlying focal epilepsy. *Neuron* 96, 401
79. Gerstner, J.R. *et al.* (2014) BMAL1 controls the diurnal rhythm and set point for electrical seizure threshold in mice. *Front. Syst. Neurosci.* 8, 121
80. Lucey, B.P. *et al.* (2015) A new model to study sleep deprivation-induced seizure. *Sleep* 38, 777–785
81. Cohen, H.B. and Dement, W.C. (1970) Prolonged tonic convulsions in REM deprived mice. *Brain Res.* 22, 421–422
82. Fang, P.C. *et al.* (2008) Seizure precipitants in children with intractable epilepsy. *Brain and Development* 30, 527–532
83. Rajna, P. and Veres, J. (1993) Correlations between night sleep duration and seizure frequency in temporal lobe epilepsy. *Epilepsia* 34, 574–579
84. Mohammed, H.S. *et al.* (2011) Neurochemical and electrophysiological changes induced by paradoxical sleep deprivation in rats. *Behav. Brain Res.* 225, 39–46
85. Yegnanarayan, R. *et al.* (2006) Chronotherapeutic dose schedule of phenytoin and carbamazepine in epileptic patients. *Chronobiol. Int.* 23, 1035–1046
86. Ramgopal, S. *et al.* (2013) Chronopharmacology of anti-convulsive therapy. *Curr. Neurol. Neurosci. Rep.* 13, 2
87. Guilhoto, L.M. *et al.* (2011) Higher evening antiepileptic drug dose for nocturnal and early-morning seizures. *Epilepsy Behav.* 20, 334–337
88. Mattis, J. and Sehgal, A. (2016) Circadian rhythms, sleep, and disorders of aging. *Trends Endocrinol. Metab.* 27, 192–203
89. Agorastos, A. *et al.* (2014) Circadian rhythmicity, variability and correlation of interleukin-6 levels in plasma and cerebrospinal fluid of healthy men. *Psychoneuroendocrinology* 44, 71–82
90. Banks, W.A. *et al.* (1994) Penetration of interleukin-6 across the murine blood-brain barrier. *Neurosci. Lett.* 179, 53–56